

Lecture 16: Stochastic processes in probabilistic modeling

Last lecture closed with Brownian motion, which is a **stochastic process**, i.e. a collection of random variables indexed by some set. Today we look at two other stochastic processes used in qualitatively different ways:

- **As likelihoods.** The data themselves are a random set of points $\{x_i\}_{i=1}^n$ — arrival times, spike trains, tree locations. The process plays the role of $p_\theta(x)$.
- **As priors.** The latent object has uncertain dimension — a mixture with an unknown number of clusters, a latent-feature model with an unknown number of features. The process plays the role of $p(\theta)$.

Poisson point processes

A **Poisson process** is a stochastic process that generates a random set of points $\{x_i\}_{i=1}^n \sim \text{PP}(\lambda(\cdot))$.

- Each point takes values in some space $x_i \in \mathcal{X}$.
- It is governed by an **intensity function** $\lambda(x) : \mathcal{X} \rightarrow \mathbb{R}_+$.
- Let $N(\mathcal{A}) \equiv \#\{i : x_i \in \mathcal{A}\}$ count the points in a measurable set $\mathcal{A} \subset \mathcal{X}$.
- Let $\Lambda(\mathcal{A}) \equiv \int_{\mathcal{A}} \lambda(x) dx$ be the total rate assigned to set $\mathcal{A}$.
- Then two properties define PPs:
 - **Property #1 (Poisson counts).** The number of points in any set is Poisson-distributed:

$$N(\mathcal{A}) \sim \text{Po}(\Lambda(\mathcal{A})).$$

- **Property #2 (Independence).** Counts on disjoint sets are independent,

$$N(\mathcal{A}) \perp N(\mathcal{B}) \iff \mathcal{A} \cap \mathcal{B} = \emptyset.$$

[Figure to insert] *Inhomogeneous Poisson process on $[0, T]$: top panel a smooth intensity curve $\lambda(x)$ with a bump or two; bottom panel tick marks at sampled event locations x_i , denser where λ is high.*

Example applications

Poisson processes are used to model many phenomena, including:

- Neural spike trains: each $x_i \in \mathbb{R}$ is the time at which a neuron fires.
- Species distributions: each $x_i \in \mathbb{R}^2$ is the location of a species.
- Cosmology: each $x_i \in \mathbb{R}^4$ is location and time of a super nova.
- Arrival times of customers in a queue, or HTTP requests to a server.
- Times of photon detections on a light sensor.

Likelihood

Given a configuration $\{x_i\}_{i=1}^n \subset \mathcal{X}$, the Poisson process likelihood is

$$p(\{x_i\}_{i=1}^n \mid \lambda) = \exp(-\Lambda(\mathcal{X})) \prod_{i=1}^n \lambda(x_i).$$

- The product $\prod_i \lambda(x_i)$ then contributes the density at each observed location.
- The exponential factor is the probability that *no* point falls anywhere else in $\mathcal{X}$.

Some simple algebra reveals the following:

$$\underbrace{\frac{\Lambda(\mathcal{X})^n e^{-\Lambda(\mathcal{X})}}{n!}}_{\text{Pois}(n; \Lambda(\mathcal{X}))} \cdot \underbrace{\prod_{i=1}^n \frac{\lambda(x_i)}{\Lambda(\mathcal{X})}}_{\text{iid locations}} = \frac{1}{n!} \exp\left(-\int_{\mathcal{X}} \lambda(x) dx\right) \prod_{i=1}^n \lambda(x_i),$$

which matches up to a factor of $1/n!$ the PP likelihood.

- Question: Why does the $\frac{1}{n!}$ represent?
- Answer: The factor for treating $\{x_i\}$ as an unordered vs ordered set.

Top-down sampling

The derivation above reveals one way to think of sampling from a PP:

Top down sampling:

1. **Sample the count.** $n \sim \text{Pois}(\Lambda(\mathcal{X}))$.
2. **Sample the locations.** $x_i \stackrel{\text{iid}}{\sim} \lambda(x)/\Lambda(\mathcal{X})$ for $i = 1, \dots, n$.

This requires the ability to compute the integral and to sample from the normalized intensity.

Intervals of a homogeneous Poisson process

A Poisson process is **homogeneous** if its intensity is constant, $\lambda(x) \equiv \lambda$.

On a time interval $[0, T] \subset \mathbb{R}$:

- **Property #3 (Exponential intervals).** Inter-arrival times are i.i.d. exponential,

$$\Delta_i = x_i - x_{i-1} \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda).$$

- **Property #4 (Memorylessness).** The time until the next point is independent of the time elapsed since the last one.

Interval sampling:

- Initialize $\{x\} = \{\}$ and $x_0 = 0$.
- For $i = 1, 2, \dots$:
 - $\Delta_i \sim \text{Exp}(\lambda)$
 - $x_i = x_{i-1} + \Delta_i$
 - if $x_i > T$: return $\{x\}$
 - else: $\{x\} = \{x\} \cup x_i$, continue

Superposition and thinning

- **Property #5 (Superposition).** The union of independent Poisson processes is a Poisson process,

$$\{x_i\}_{i=1}^n \sim \text{PP}(\lambda_1(x)) \text{ and } \{x'_j\}_{j=1}^m \sim \text{PP}(\lambda_2(x)) \implies \{x_i\} \cup \{x'_j\} \sim \text{PP}(\lambda_1 + \lambda_2).$$

- **Property #6 (Poisson thinning).** Given $\{x_i\} \sim \text{PP}(\lambda_1 + \lambda_2)$, draw an independent label for each point,

$$z_i \sim \text{Bern}\left(\frac{\lambda_1(x_i)}{\lambda_1(x_i) + \lambda_2(x_i)}\right).$$

Then $\{x_i : z_i = 1\}$ and $\{x_i : z_i = 0\}$ are independent Poisson processes with intensities λ_1 and λ_2 . (This generalizes to more than two processes.)

Cox processes

A Cox process (or **doubly stochastic Poisson process**) further assumes the intensity is random, $\lambda \sim p(\lambda)$. There are different kinds of priors we place over the function λ .

A **Neyman–Scott process** is one option. Here we convolve a **latent** Poisson process with a non-negative kernel.

- Sample latent "parent" locations $\{c_k\} \sim \text{PP}(G(\cdot))$ (often homogeneous, $G(x) \equiv G$).
- Define a non-negative kernel $k(x - c_k) \geq 0$
- Often Gaussian $k(x - c_k) = \mathcal{N}(x; c_k, \sigma^2)$
- Sample a Poisson process with intensity: $\lambda(x) = \alpha \sum_k k(x - c_k)$
- $\alpha > 0$ controls the expected number of "offspring" per parent

[INSERT PICTURE]

- Inference is **hard** because the latent parents must be marginalized.
- Standard approaches treat $\{c_k\}$ as latent variables and run EM or MCMC.

A **log-Gaussian Cox process** is another option. Here we place a **Gaussian process prior** on $\log \lambda(x) \sim \text{GP}(m, k)$.

- A **Gaussian process** $f \sim \text{GP}(m, K)$ is a distribution over functions $f : \mathcal{X} \rightarrow \mathbb{R}$
- Any finite collection $(f(x_1), \dots, f(x_k))$ is then jointly Gaussian
- Fully specified by a mean function $m(x)$ and a covariance kernel $k(x, x')$.

$$(f(x_1), \dots, f(x_k)) \sim \mathcal{N}(\mathbf{m}, \mathbf{K}), \quad \mathbf{m}_i = m(x_i), \quad \mathbf{K}_{ij} = k(x_i, x_j).$$

- Setting $\lambda(x) = e^{f(x)}$ gives a flexible, strictly-positive intensity whose smoothness is controlled by K .

[INSERT PICTURE]

- Inference is **hard** because the log-likelihood contains the integral $\int_{\mathcal{X}} e^{f(x)} dx$, which depends on f everywhere.
- One option: **discretize** $\mathcal{X}$ into cells:
 - Each cell count is $N(\mathcal{A}_m) \sim \text{Po}(|\mathcal{A}_m| e^{f(c_m)})$ at a cell center c_m .
 - The GP prior collapses to a **multivariate Gaussian prior** on the cell log-rates
 - Reduces to a finite-dimensional (non-conjugate) Poisson model with MVN prior
- Or use **variational inference** with sparse GP approximations.
 - Here we choose $M \ll n$ **inducing locations** $\mathbf{z} = (z_1, \dots, z_M)$.

- Define the inducing values $\mathbf{u} = (f(z_1), \dots, f(z_M))$
- Place a Gaussian variational distribution over them:

$$q(\mathbf{u}) = \mathcal{N}(\mathbf{m}_u, \mathbf{S})$$

- Variational parameters $\mathbf{m}_u \in \mathbb{R}^M$ and $\mathbf{S} \in \mathbb{R}^{M \times M}$.
- Sampling $f \sim q$ is then two-step:
 - $\mathbf{u} \sim \mathcal{N}(\mathbf{m}_u, \mathbf{S})$
 - $f(x) \mid \mathbf{u} \sim \mathcal{N}(\mathbf{k}_x^\top \mathbf{K}_{uu}^{-1} \mathbf{u}, k(x, x) - \mathbf{k}_x^\top \mathbf{K}_{uu}^{-1} \mathbf{k}_x)$,
 - $\mathbf{k}_x = (k(x, z_1), \dots, k(x, z_M))^\top$
 - $\mathbf{K}_{uu}$ is the $M \times M$ kernel matrix at the inducing points
- Or various **augmentation schemes** with auxiliary variables [...]

Renewal processes

Exponentially-distributed arrivals is a good assumption for "pure chance" events (e.g., lightning strikes, horse-kick deaths). However, in many applications, the timing of arrivals may exhibit either more "clumpiness" or more "regularity".

- Example of **"clumpiness"**:
 - News articles about some conflict
 - Locations of colony-based insects
- Example of **"regularity"**:
 - Timing of subscription-based purchases
 - Locations of trees

One way to generalize temporal Poisson processes is via **renewal processes**:

$$\Delta_i = x_i - x_{i-1} \sim f_\theta(\Delta)$$

where f_θ is any distribution over $\mathbb{R}_{\geq 0}$. An example is the **gamma renewal process**:

$$\Delta_i = x_i - x_{i-1} \sim \text{Gamma}(k, k\lambda)$$

where if $k = 1$ we get a Poisson process but if $k < 1$ we get more clumpy arrivals and $k > 1$ we get more regular arrivals.

[INSERT PICTURE]

Hawkes processes

A disadvantage of Poisson point processes is that they do not capture any dependence between past generated events and future ones. **Autoregressive point processes** redefines the intensity function (in continuous time) a function of the history (or "filtration") $\mathcal{H}_t \equiv \{x_i : x_i < t\}$:

$$\lambda(t \mid \mathcal{H}_t)$$

One example is the **Hawkes process** which defines the rate function as:

$$\lambda(t \mid \mathcal{H}_t) = \lambda_0 + \sum_{x_i \in \mathcal{H}_t} h(t - x_i),$$

- $\lambda_0 \in \mathbb{R}_+$ is a **baseline rate**
- $h : \mathbb{R}_+ \mapsto \mathbb{R}_+$ is a positive **impulse response** (or **influence function**).
- Past events "excite" future ones.
- Good model for bursty spike trains, financial trade arrivals, social-media cascades, and earthquake aftershocks...

A common parametric choice is the **exponential impulse response**,

$$h(\Delta t) = \frac{w}{\tau} e^{-\Delta t/\tau} = w \cdot \text{Exp}(\Delta t; \tau),$$

- $\tau \in \mathbb{R}_+$ is a time constant governing the rate of decay
- $w \in \mathbb{R}_+$ is a scaling parameter such that $\int_0^\infty h(\Delta t) d\Delta t = w$.
- Intuitively, w is the expected number of "offspring" events triggered by each parent event.

Branching-process view.

- By Poisson superposition, the Hawkes process is equivalent to a **branching process**.
- An initial background Poisson process with rate λ_0 generates "root" events.
- Each event x_i independently spawns its own Poisson process of offspring with intensity $h(t - x_i)$.
- The full Hawkes process is then the union of the background and all offspring processes.
- Since each event has on average w direct offspring, the process is **stable** (finite cluster sizes) when $w < 1$.

[INSERT PICTURE]

MLE for Hawkes processes. Suppose we observe $\{x_i\}_{i=1}^n \subset [0, T]$ and want to estimate the parameters $\theta = (\lambda_0, w)$ of a Hawkes process with an exponential impulse response (consider τ to be fixed). The Hawkes log-likelihood has the same form as that of a Poisson process,

$$\log p(\{x_i\}_{i=1}^n \mid \theta) = - \int_0^T \lambda_\theta(t \mid \mathcal{H}_t) dt + \sum_{i=1}^n \log \lambda_\theta(x_i \mid \mathcal{H}_{x_i}).$$

Substituting the conditional intensity and approximating the integral (using $\int_0^T \text{Exp}(t - x_i; \tau) dt \approx 1$ when $T - x_i \gg \tau$), this simplifies to

$$\log p(\{x_i\}_{i=1}^n \mid \theta) \approx -\theta^\top \phi_0 + \sum_{i=1}^n \log(\theta^\top \phi_i),$$

where $\phi_0 = (T, n)^\top$ and $\phi_i = (1, \sum_{x_j \in \mathcal{H}_{x_i}} \text{Exp}(x_i - x_j; \tau))^\top$. This is a sum of logs of linear forms in θ — concave, and so easily maximized by standard convex optimization.

From data models to priors over structure

We've been treating stochastic processes as models of random points. Now use the same machinery to build **priors over random discrete measures** — distributions on the space of measures.

Motivating problem: a mixture model with K components, where we don't want to fix K up front.

The route:

1. Write down the finite- K Bayesian mixture.
2. Derive its collapsed Gibbs sampler.
3. Send $K \rightarrow \infty$.
4. Read off the implicit prior $\Rightarrow$ Dirichlet process.

A finite Bayesian mixture

A finite Bayesian mixture model with K components is defined by the following generative process:

1. Sample mixing proportions from a Dirichlet prior,

$$\boldsymbol{\pi} \sim \text{Dirichlet}(\boldsymbol{\alpha}).$$

2. Sample component parameters,

$$\boldsymbol{\theta}_k \stackrel{\text{iid}}{\sim} H, \quad k = 1, \dots, K.$$

3. Sample assignments and observations,

$$z_i \stackrel{\text{iid}}{\sim} \boldsymbol{\pi}, \quad x_i \mid z_i, \boldsymbol{\theta} \sim F(\boldsymbol{\theta}_{z_i}), \quad i = 1, \dots, n.$$

- We will set $\alpha = (\alpha/K) \mathbf{1}_K$, so the *total* prior mass on π is α , independent of K .
- This is the choice that makes the $K \rightarrow \infty$ limit well-behaved.

Collapsed Gibbs sampler

- We can marginalize out π by Dirichlet–multinomial conjugacy.
- When H is conjugate to F , we can also marginalize out θ_k .
- Define $n_k^{(-i)} = \#\{i' \neq i : z_{i'} = k\}$
- The complete conditional for z_i is then

$$p(z_i = k \mid \mathbf{z}_{-i}, \mathbf{x}) \propto \underbrace{(\alpha/K + n_k^{(-i)})}_{\text{cluster size}} \cdot \underbrace{p(x_i \mid \{x_{i'} : z_{i'} = k\}, H)}_{\text{posterior predictive}},$$

- Notice that z_i depends on $\mathbf{z}_{-i}$.
- The posterior predictive measures how well x_i fits the other points in cluster k , and is closed-form in the conjugate case.
- Larger clusters attract new points.

Letting $K \rightarrow \infty$

Look at the prior factor $\alpha/K + n_k^{(-i)}$ as $K \rightarrow \infty$.

- **Occupied cluster** ($n_k^{(-i)} \geq 1$): prior weight $\rightarrow n_k^{(-i)}$.
- **Empty cluster**: prior weight $\alpha/K \rightarrow 0$ individually, but the *total* prior on all (infinitely many) empty clusters stays at α .

So the conditional becomes

$$p(z_i = k \mid \mathbf{z}_{-i}, \mathbf{x}) \propto \begin{cases} n_k^{(-i)} \cdot p(x_i \mid \{x_{i'} : z_{i'} = k\}, H) & \text{if } k \text{ is occupied} \\ \alpha \cdot p(x_i \mid H) & \text{if } k \text{ is new.} \end{cases}$$

- **This is a complete, well-defined (collapsed) Gibbs sampler.**
- Notice it never instantiates the infinite-dimensional π or the infinite collection of θ_k 's.
- It only tracks occupied clusters and their sufficient stats.
- New clusters are sampled and old ones are deleted (when empty).
- K is never fixed; the number of occupied clusters can grow or shrink at each step.
- Note that the **effective** $K \leq n$ at each sampling iteration

Infinite- K collapsed Gibbs (Algorithm 3 of Neal 2000):

```
initialize  $z_1, \dots, z_n$  (e.g. all in one cluster)
until convergence:
  for  $i = 1, \dots, n$ :
    remove  $x_i$  from its cluster; delete cluster if it becomes empty
    for each currently occupied cluster  $k$ :
       $\text{weight}_k = n_k^{(-i)} * p(x_i | \{x_{i'} : z_{i'} = k\}, H)$ 
     $\text{weight}_{\text{new}} = \alpha * p(x_i | H)$ 
    sample  $z_i \sim \text{Categorical}([\text{weights}, \text{weight}_{\text{new}}] / \text{total})$ 
    if  $z_i$  is the "new" outcome:
      spawn a new cluster (draw  $\theta_{\text{new}} \sim H$  if maintaining explicit params)
      add  $x_i$  to its (new) cluster
```

Three views: CRP, DP, stick-breaking

The infinite- K sampler is an inference algorithm for *some* model. What prior did we just commit to?
Three equivalent angles.

Chinese Restaurant Process (CRP)

- One answer is that our prior was what's called the **Chinese Restaurant Process** (CRP)
- Imagine n customers entering a restaurant one at a time.
- Customer i , given the previous $i - 1$ already seated, joins table k with probability proportional to its size
- ...or starts a new table with probability proportional to α :

$$p(z_i = k \mid z_{1:i-1}) = \begin{cases} \frac{n_k}{i - 1 + \alpha} & \text{occupied table } k \\ \frac{\alpha}{i - 1 + \alpha} & \text{a new table.} \end{cases}$$

- This defines a stochastic process prior over $p(z_1, z_2, \dots \mid \alpha)$
- It implicitly defines a prior over K
- Note that the joint over assignments is **exchangeable**
- Even though the rule looks sequential, labels i can be permuted; it induces the same distribution over **partitions**

[Figure to insert] Round-table picture of a Chinese restaurant: a row of occupied tables with seated customers, plus one empty table off to the side. Arrows from "next customer" show

relative probabilities of joining each table ($\propto n_k$) versus starting a new one ($\propto \alpha$).

Dirichlet process (DP)

- The CRP arises as the marginal over partitions of a more fundamental object.
- Define a **discrete measure** G on Θ .

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}$$

- The atoms θ_k^* are points in Θ
- The measure of a subset is $\mathcal{A} \subset \Theta$:

$$G(\mathcal{A}) = \sum_{k=1}^{\infty} \pi_k \delta(\theta_k^* \in \mathcal{A})$$

- The weights $\pi_k \geq 0$ sum to 1.
- Thus $\sum_j G(\mathcal{A}_j) = 1$ for any partition $(\mathcal{A}_j)_j$ of Θ
- A **random discrete measure** G is now one where the weights π_k and atoms θ_k^* are random.
- A **Dirichlet process** is a stochastic process over discrete random measures:

$$G \sim \text{DP}(\alpha, H),$$

- The **base measure** H over Θ is the distribution over atom locations: $\theta_k^* \stackrel{\text{iid}}{\sim} H$
- The **concentration** $\alpha > 0$ governs how many distinct atoms will have non-negligible probability.
- Property: for every finite measurable partition $(\mathcal{A}_1, \dots, \mathcal{A}_K)$ of Θ ,

$$(G(\mathcal{A}_1), \dots, G(\mathcal{A}_K)) \sim \text{Dir}(\alpha H(\mathcal{A}_1), \dots, \alpha H(\mathcal{A}_K)).$$

- Now consider drawing a unique parameter for every data point i :

$$\theta_i \mid G \stackrel{\text{iid}}{\sim} G, \quad i = 1, \dots, n.$$

- Note that this looks different than before, when we draw a cluster assignment z_i .
- However, since G is a discrete measure, there is a **non-negligible probability of ties**:

$$p(\theta_i = \theta_{i'}) > 0$$

- So we can redefine z_i as recording which atom θ_k^* equals θ_i .
- If we now marginalize out G , we recover the CRP over $\mathbf{z}$.
- **Conjugacy.** Given $\theta_{1:n} \mid G \stackrel{\text{iid}}{\sim} G$, the posterior is again DP:

$$G \mid \theta_{1:n} \sim \text{DP}\left(\alpha + n, \frac{\alpha}{\alpha+n}H + \frac{1}{\alpha+n} \sum_i \delta_{\theta_i}\right).$$

The **DP mixture model** stacks a likelihood on top:

$$G \sim \text{DP}(\alpha, H), \quad \theta_i \mid G \stackrel{\text{iid}}{\sim} G, \quad x_i \mid \theta_i \sim F(\theta_i).$$

- Marginalizing G and recording the ties among θ_i 's gives the CRP
- Our collapsed Gibbs sampler (above) is doing inference in exactly this model

Stick-breaking (Sethuraman 1994)

An explicit constructive description of $G \sim \text{DP}(\alpha, H)$:

$$V_k \stackrel{\text{iid}}{\sim} \text{Beta}(1, \alpha), \quad \pi_k = V_k \prod_{j < k} (1 - V_j), \quad \theta_k \stackrel{\text{iid}}{\sim} H,$$

- This is a way to generate $G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k}$.
- Picture: start with a unit-length stick.
 - Break off a fraction V_1 for component 1
 - Break a fraction V_2 of what's left for component 2, and so on.
 - Smaller α puts more mass on the first few atoms.

[Figure to insert] *Stick-breaking diagram: unit-length horizontal stick at top, broken at V_1 to give π_1 ; remainder $(1 - V_1)$ shown below, broken at V_2 to give π_2 ; successively smaller pieces.*

Label $V_k \stackrel{\text{iid}}{\sim} \text{Beta}(1, \alpha)$.

- Stick-breaking is the most useful construction for **variational** inference: truncate at $K_{\max}$ and run mean-field updates.
- Three samplers fit the DP mixture:
 - collapsed Gibbs on the partition (Neal Algorithm 3, above)
 - truncated stick-breaking with CAVI/SVI
 - Slice sampling (Walker 2007).
- Same model, different trade-offs.

Completely random measures

- The DP is one instance of a **random discrete measure** $G = \sum_k w_k \delta_{\theta_k}$.
- We built it as a $K \rightarrow \infty$ limit on normalized Dirichlet weights.

- There is a more general, more elegant picture: **build G from a Poisson process on the product space (parameter axis) $\times$ (weight axis).**

Poisson random measures

Construct G by sampling weight–location pairs from a **marked Poisson process** on the product space,

$$\{(\boldsymbol{\theta}_k, w_k)\}_{k=1}^{\infty} \sim \text{PP}(\nu) \quad \text{on } \Theta \times \mathbb{R}_+, \quad G = \sum_k w_k \delta_{\boldsymbol{\theta}_k}.$$

- The intensity function $\nu(d\boldsymbol{\theta}, dw)$ is called the **Lévy intensity** of the resulting measure.
- The random measures built this way are exactly the **completely random measures (CRMs)** of Kingman (1967)
- Characterized by the property that $G(\mathcal{A}_1), \dots, G(\mathcal{A}_k)$ are independent for disjoint $\mathcal{A}_1, \dots, \mathcal{A}_k \subset \Theta$.
- Different choices of ν give different priors.

[Figure to insert] _3D picture: horizontal axis Θ (parameter space), depth axis $\mathbb{R}_+$ (weight axis). Scatter of dots in the base plane = one Poisson process realization, each dot $(\boldsymbol{\theta}_k, w_k)$ = one atom. Vertical sticks of height w_k from each base location = stem plot of G on Θ . Translucent surface above the base plane hints at the Lévy intensity ν .

Gamma process and Dirichlet process

Choose the Lévy intensity

$$\nu_{\Gamma}(d\boldsymbol{\theta}, dw) = \alpha H(d\boldsymbol{\theta}) \cdot w^{-1} e^{-w} dw.$$

- The w^{-1} factor is improper at zero; this produces a countably infinite number of atoms.
- $G(\Theta) \sim \text{Gamma}(\alpha, 1)$ almost surely.
- More generally, for any finite measurable partition $(\mathcal{A}_1, \dots, \mathcal{A}_k)$ of Θ , the masses are **independent gammas**:

$$G(\mathcal{A}_j) \stackrel{\text{ind}}{\sim} \text{Gamma}(\alpha H(\mathcal{A}_j), 1), \quad j = 1, \dots, k.$$

- This is the **gamma process**, $G \sim \text{GaP}(\alpha, H)$.
- Although infinitely many atoms exist, only **finitely many have non-negligible weight**.
- By the Poisson process property #1, the count of atoms with w_k above any threshold $\epsilon > 0$ is itself Poisson:

$$\#\{k : w_k > \epsilon\} \sim \text{Po}\left(\alpha \int_{\epsilon}^{\infty} w^{-1} e^{-w} dw\right),$$

- The Poisson rate is finite for every $\epsilon > 0$.
- As $\epsilon \rightarrow 0$ the rate diverges like $\alpha \log(1/\epsilon)$, recovering the infinite cloud in the limit.
- So the gamma process effectively behaves like a random discrete measure with a **finite** but random number of visible atoms.
- Normalizing by the total recovers the **Dirichlet** finite-partition law of the DP.

$$G \sim \text{GaP}(\alpha, H) \implies G \equiv G/G(\Theta) \sim \text{DP}(\alpha, H).$$

- **Dirichlet processes are normalized gamma processes.**
- The DP throws away one degree of freedom (the total mass) by normalizing.

Alternative route: $K \rightarrow \infty$ on unnormalized weights

- If $w_1, \dots, w_K \stackrel{\text{iid}}{\sim} \text{Gamma}(\alpha/K, 1)$ and $S = \sum_k w_k$, then $(w_1/S, \dots, w_K/S) \sim \text{Dir}(\alpha/K, \dots, \alpha/K)$ and is independent of $S \sim \text{Gamma}(\alpha, 1)$.
- Sending $K \rightarrow \infty$ on the unnormalized w_k 's, each one concentrates near zero with a heavy right tail, total mass preserved — gives exactly the gamma process. The K-limit gives the picture; the Lévy form unlocks the broader CRM family.

Beta process and the Indian Buffet Process

Beta process

Choose a Lévy intensity that confines weights to $[0, 1]$:

$$\nu_B(d\theta, dw) = \alpha H(d\theta) \cdot w^{-1}(1-w)^{\alpha-1} dw.$$

- The w^{-1} factor is improper at zero — again a countably infinite cloud of atoms with mostly small weights.
- The $(1-w)^{\alpha-1}$ tail pushes atoms away from 1; most weights live near 0.
- The resulting $G = \sum_k w_k \delta_{\theta_k}$ has weights $w_k \in [0, 1]$, naturally interpreted as **Bernoulli probabilities**.
- For each data point x_i , draw $z_{ik} \sim \text{Bern}(w_k)$ at every atom θ_k .
- Each data point gets a (random, finite) *subset* of features — appropriate when items have overlapping attributes rather than category membership.
- This is the **beta process**, $G \sim \text{BeP}(\alpha, H)$.

Indian Buffet Process (IBP)

Marginalize G out of the beta–Bernoulli model and the binary feature matrix $Z = (z_{ik})$ has an exchangeable distribution on $n \times \infty$ binary matrices — the **Indian Buffet Process** (Griffiths & Ghahramani 2011).

- The marginal seating rule is sequential, like the CRP.
- Customer i tries each previously-tasted dish k with probability m_k/i , where m_k counts prior customers using dish k .
- Customer i then tries $\text{Po}(\alpha/i)$ brand-new dishes.
- The expected number of features used by n data points grows like $\alpha \log n$ — sublinear, controlled by α .

Indian Buffet Process – sample an $n \times \infty$ binary feature matrix Z :

```
Z = [] # row i is data point i's feature vector
K_used = 0 # number of distinct features used so far
for i = 1, ..., n:
    row_i = zeros(K_used)
    for k = 1, ..., K_used:
        m_k = sum of Z[:, k] # number of prior customers using feature k
        row_i[k] = Bernoulli(m_k / i)
    k_new = Poisson( $\alpha / i$ ) # new features sampled by customer i
    row_i = concat(row_i, ones(k_new))
    Z = pad_columns(Z, k_new); append row_i to Z
    K_used += k_new
return Z
```

[Figure to insert] _Small binary feature matrix Z (say 10 rows, columns sorted by descending column sum). Black squares for $z_{ik} = 1$, white for 0. Drop-off from filled cells on the left (popular dishes) to sparse cells on the right (rarely-tasted) captures the IBP's heavy left-skew._